

A Stochastic Dynamic Programming Analysis of Farmland Investment and Financial Management

Heman D. Lohano¹ and Robert P. King²

¹Department of Economics and Finance, Institute of Business Administration, Main Campus, University Road, Karachi 75270, Pakistan (corresponding author: phone: 92-300-304-2034; fax: 92-21-921-5528; e-mail: loha0002@ummn.edu).

²Department of Applied Economics, University of Minnesota, 1994 Buford Avenue, St. Paul, MN (phone: 612-625-1273; fax: 612-625-6245; e-mail: rking@umn.edu).

This paper develops a multiperiod investment portfolio model that includes risky farmland, risky and risk-free nonfarm assets, and debt financing on farmland in the presence of transaction costs and credit constraints. The model is formulated as a stochastic continuous-state dynamic programming problem, and is solved numerically for Southwestern Minnesota, USA. Results show that optimal investment decisions are dynamic and take into account the future decisions due to uncertainty, partial irreversibility, and the option to wait. The optimal policy includes ranges of inaction, states where the optimal policy in the current year is to wait. The risk-averse farmer makes a lower investment in risky farmland reflecting risk-avoiding behavior. We find that, in addition to risk aversion, the length of the planning horizon affects risk-avoiding behavior in investment decisions. In contrast to a static model, changes in the riskiness of returns affect optimal investment decisions even when the decision maker is risk neutral. Finally, we find that higher debt financing on farmland is optimal when risky nonfarm assets can be included in the optimal investment portfolio and that the probability of exiting farming increases with the risky nonfarm investment.

Dans le présent article, nous avons élaboré un modèle de portefeuille multipériode composé d'actifs agricoles risqués (terres), d'actifs non agricoles risqués et sans risque ainsi que de financement par emprunt de terres comprenant des coûts de transaction et des contraintes de crédit. Le modèle a été formulé comme un problème de programmation dynamique stochastique en temps continu et a été résolu de façon numérique pour le sud-ouest du Minnesota, aux États-Unis. Les résultats ont montré que les décisions optimales d'investissement sont dynamiques et qu'elles tiennent compte des décisions futures en raison de l'incertitude, de l'irréversibilité partielle et du choix d'attendre. La politique optimale inclut des plages d'inaction et des situations où la politique optimale de l'année en cours doit attendre. L'agriculteur risquophobe investit peu dans des actifs fonciers risqués ce qui traduit un comportement d'évitement des risques. Nous sommes arrivés à la conclusion que la longueur de l'horizon de planification, en plus de l'aversion pour le risque, influe sur le comportement d'évitement des risques au moment de prendre des décisions d'investissement. Contrairement au modèle statique, les changements dans les risques de rendements influent sur les décisions optimales d'investissement même si le décideur est indifférent au risque. Finalement, nous avons trouvé que le financement par emprunt pour des terres agricoles est optimal lorsqu'il est possible d'inclure des actifs non agricoles risqués dans le portefeuille optimal et que la probabilité de quitter l'agriculture augmente en présence d'investissements non agricoles risqués.

INTRODUCTION

The agricultural sector of the United States has experienced substantial change in farm structure over time. From 1950 to 2007, average farm size grew from 213 to 449 acres, and the number of farms declined from 5.6 to 2.1 million (USDA 2008). This pattern of change is associated with the growth of successful farms along with the failure of some farms to respond optimally to increasing market and financial risks (Baker 1977). This process of change underscores the need to investigate the farm firm growth problem centering on farmland investment decisions in the presence of debt financing.

Farmland is a risky investment. In addition to risk in farm returns, farmers also face farmland value risks. Farmland investments are usually financed using a combination of equity and debt capital, and a firm's decision about debt capital is of major importance. Since farmland is collateral, low farmland values decrease the maximum available debt capital. In periods of low farm income, with an associated decline in land values, farmers experience financial stress, which results from their inability to meet planned cash flow commitments (Brake 1983). During periods of financial stress, as in the early 1980s, farms with high debt-to-asset ratios face a higher risk of failure (Jensen and Langemeier 1996). Thus, in addition to explaining the changing structure of agriculture, the analysis of farmland investment with debt financing addresses financial management and farm survival issues.

The literature on investment under uncertainty emphasizes the need for firms to consider future decisions and uncertainty when making current decisions (Dixit and Pindyck 1994). Dynamic programming (DP) is a flexible tool that accommodates the time relationship of decisions and outcomes in a sequential decision process under uncertainty (Puterman 1989; Rust 1996). Relatively few studies have analyzed farmland investment decisions in a stochastic and dynamic framework. Minden (1968) formulated a conceptual farm firm growth model in a DP framework. Larson et al (1974) developed an empirical DP model of farmland investment and financial decisions. Their model incorporated transaction costs of selling farmland and uncertainty of crop yield with deterministic farmland values. Schnitkey et al (1989) extended the model by incorporating dynamics in stochastic farm returns and land prices. They estimated the relationship between land price and farm returns but treated this as deterministic in their DP model. However, this replaces a random variable with its expected value, and could lead to erroneous results in a general DP model (Nemhauser 1966, p. 156). Moreover, when land values are assumed to be deterministic, a large component of risk is not considered in evaluating the survivability of farms and the profitability of farmland purchase and sale decisions. Land value risk is particularly significant in investment decisions involving collateral risks and credit reserve risks, since land value serves as collateral in debt financing (Featherstone et al 1990). Incorporating land value risk is also important for full consideration of capital gain or loss risks. Further, Pederson and Brake (1982), and others, have recognized that lenders limit the proportion of debt capital for financing farmland investment. Thus, it is important to incorporate the credit constraint in a DP model of farmland investment and financial decisions.

Other factors also affect firm growth decisions. Besides farmland investment, farmers have various nonfarm investment opportunities, such as stocks, bonds, and mutual funds. Given numerous uncertainties in farming, it is important to investigate the optimal mix

of farmland and nonfarm assets and to explore its impact on farm financial structure and growth. Risk aversion is a well-known factor affecting investment and financial strategies, but risk aversion alone cannot explain the large difference in returns on stocks and Treasury bills (Kocherlakota 1996; Chavas and Thomas 1999). For example, Collins and Karp (1993) indicate that the optimal choice of leverage is also affected by a farmer's age. There is a need, then, to explore how other factors—including nonfarm investment opportunities, risk aversion, length of planning horizon—affect farmers' investment and financial strategies.

This paper develops a finite horizon multiperiod investment portfolio model that includes risky farmland, risky and risk-free nonfarm assets, and debt financing on farmland in the presence of transaction costs and credit constraints. The model is formulated as a stochastic continuous-state DP problem and is solved numerically to determine optimal investment and financial strategies for Southwestern Minnesota farms. We analyze the effect of interest rate, length of the planning horizon, riskiness of farm returns, risk preferences, and risky nonfarm assets on farmland investment and financial management decisions.

Numerically solving a large-scale stochastic DP model presents computational and technical challenges. The DP algorithm requires optimization and integration. State variables may go out of bounds, and the utility function needs to be defined for all values of states. Our description of procedures used to solve the investment problem developed for this study provides insights on these issues that can be applied in other settings.

MODEL

Consider the case where a farm manager makes decisions on levels of investment in risky farmland, risky and risk-free nonfarm assets, and debt financing on farmland at the beginning of each year over a finite planning horizon. The farm manager's objective is to maximize expected utility of terminal net wealth. We formally specify this problem as follows:

$$\max_{\{(x_t, x_{mt})\}_{t=0}^T} E_0[u(W_{T+1})] \quad (1)$$

subject to

$$\ln R_{t+1} = \beta_0 + \beta_1 \ln R_t + \varepsilon_{1,t+1} \quad (2)$$

$$\ln P_{t+1} = \alpha_0 + \alpha_1 \ln P_t + \alpha_2 \ln R_t + \varepsilon_{2,t+1} \quad (3)$$

$$\ln M_{t+1} = \gamma_0 + \varepsilon_{3,t+1} \quad (4)$$

$$L_{t+1} = L_t + x_t \quad (5)$$

$$\begin{aligned} W_{t+1} = (1+r)[\{W_t - (P_t + \kappa - tc_s)L_t\} - (P_t + \kappa + tc)x_t - cL_{t+1} - x_{mt}] \\ + M_{t+1}x_{mt} + R_{t+1}L_{t+1} + (P_{t+1} + \kappa - tc_s)L_{t+1} \end{aligned} \quad (6)$$

$$x_t \in X_t \quad (7)$$

$$x_{mt} \in X_{mt} \quad (8)$$

for $t = 0, 1, 2, \dots, T$, and (R_0, P_0, L_0, W_0) are given

where

$$r = \begin{cases} r_b & \text{if } \{[W_t - (P_t + \kappa - tc_s)L_t] - (P_t + \kappa + tc)x_t - cL_{t+1} - x_{mt}\} < 0 \\ r_l & \text{otherwise} \end{cases} \quad (9)$$

$$tc = \begin{cases} tc_p & \text{if } x_t > 0 \\ -tc_s & \text{otherwise} \end{cases} \quad (10)$$

and the variables and parameters of the model are defined in Table 1.

Equations (2)–(4) describe stochastic processes for the three stochastic variables in the model: gross return per acre from crops (R_t), farmland price (P_t), and mutual fund return (M_t), respectively. Note that the gross return from crops captures the effects of both price and yield uncertainty. The mutual fund return is equal to 1 plus the annual rate of return from the mutual fund, which includes capital gains or losses as well as dividends. Specification and estimation of these equations are presented in the next section.

Table 1. Definition of variables and parameters

Variable/Parameter	Definition
t	Time index for the beginning of each year where $t = 0, 1, \dots, T$
R_t	Gross return per acre from crops produced for sale in year t
P_t	Farmland price per acre in year t
M_t	Mutual fund return in year t
L_t	Farmland acreage owned by the farmer in year t
W_t	Net wealth in year t
x_t	Farmland acreage purchased or sold by the farmer in year t
x_{mt}	Investment in the mutual fund measured in monetary units
κ	Value of machinery and equipment required per acre of farmland
tc_p	Total transaction costs per acre on purchasing farmland, machinery, and equipment
tc_s	Total transaction costs per acre on selling farmland, machinery, and equipment
r_b	Interest rate on borrowing (debt financing on farmland)
r_l	Interest rate on lending (risk-free investment)
c	Costs of production per acre for producing crops
X_t	Set of all levels of control variable x_t
X_{mt}	Set of all levels of control variable x_{mt}
$u(\cdot)$	Utility function
E_0	Expectation operator over the random shocks ε_{1t} , ε_{2t} , and ε_{3t}

At the beginning of each year t , the farm manager owns farmland acreage (L_t) and can purchase or sell land (x_t), as described in Equation (5). Each acre of land requires a fixed investment in machinery and equipment, denoted by κ . There are transaction costs on purchasing or selling farmland, machinery, and equipment, as defined in Equation (10). Besides farmland, the farm manager can invest in a risky nonfarm asset: a Standard and Poor's 500 (S&P 500) Stock Composite Index mutual fund measured in monetary units (x_{mt}). We assume that there are no transaction costs related to the buying and selling of mutual funds, so the amount invested in the mutual fund can be decided every year, subject to liquidity constraints, and adjusted accordingly no matter how much is already invested. Thus, the level of investment in the mutual fund for any year is fully described by the control variable x_{mt} .¹

Equation (6) describes the dynamics of net wealth (W_t). Net wealth is defined as

$$W_t \equiv A_t + (P_t + \kappa - tc_s)L_t \quad (11)$$

where A_t denotes cash balance and $(P_t + \kappa - tc_s)L_t$ is net sale-value of farmland, machinery, and equipment. Dynamics of cash balances are described as

$$A_{t+1} = (1 + r)[A_t - (P_t + \kappa + tc)x_t - cL_{t+1} - x_{mt}] + M_{t+1}x_{mt} + R_{t+1}L_{t+1} \quad (12)$$

The expression in brackets in Equation (12) is the net cash balance, which is current cash balance, A_t , minus cash flows associated with the purchase or sale of farmland, machinery, and equipment, $(P_t + \kappa + tc)x_t$, funds set aside for crop production expenses, cL_{t+1} ,² and funds allocated for mutual fund investment, x_{mt} . Net cash balance can be negative or positive, where a negative value represents a debt position that requires payment of a fixed interest rate, r_b ; a positive value represents cash balances that can be invested at a risk-free rate of return, r_f . The farm manager faces a difference between the borrowing and lending rates such that $r_l < r_b$ (Hirshleifer 1970). Cash balance in year $t + 1$ is equal to the sum of: net cash balances multiplied by 1 plus the relevant interest rate, r ; the realized cash value of mutual funds, $M_{t+1}x_{mt}$; and revenue from crop production, $R_{t+1}L_{t+1}$. Equation (6) is derived by rewriting Equation (11) for period $t + 1$ as $W_{t+1} \equiv A_{t+1} + (P_{t+1} + \kappa - tc_s)L_{t+1}$, substituting Equation (12) for A_{t+1} , and then in the result substituting $A_t \equiv W_t - (P_t + \kappa - tc_s)L_t$ for current cash balance. In Equation (6), net wealth in period $t + 1$ is the sum of cash balance in period $t + 1$, as given in Equation (12), and the net sale-value of farmland, machinery, and equipment in period $t + 1$.³

The current cash balance, A_t , can be positive or negative. In reporting results in subsequent sections we often calculate the debt-to-farm-asset ratio for given values of state variables and parameters. The debt-to-farm-asset ratio is defined as

$$DFA_t = \max \left\{ 0, \frac{-A_t}{(P_t + \kappa - tc_s)L_t} \right\} \quad (13)$$

There are four constraints on farmland purchases or sales (x_t): a feasibility constraint, a bankruptcy condition, constraint on re-entry in farming, and a credit constraint. The feasibility constraint satisfies $\underline{L} \leq L_{t+1} \leq \bar{L}$, where $\underline{L}$ denotes minimum acres required for farming and $\bar{L}$ denotes the maximum feasible acres the manager can own in the location. Substituting $L_{t+1} = L_t + x_t$ in this constraint and rearranging yields: $\underline{L} - L_t \leq$

$x_t \leq \bar{L} - L_t$. In addition, the farm manager can sell all land: $x_t = -L_t$. Thus, the feasibility constraint is represented by

$$X_{FCt} \equiv \{x_t : (L - L_t) \leq x_t \leq (\bar{L} - L_t), x_t = -L_t\} \quad (14)$$

Under the bankruptcy condition, the farm is liquidated if net wealth is negative

$$x_t = -L_t \quad \text{if } W_t < 0 \quad (15)$$

In addition, when $W_t \geq 0$, the manager can potentially choose to sell all land, as indicated in Equation (14). We assume that once all land is sold, the business does not re-enter farming. Therefore

$$\text{if } x_t = -L_t, \quad \text{then } x_{t+1} = x_{t+2} = \dots = x_T = 0 \quad (16)$$

Finally, the credit constraint limits access to loans. It restricts the level of purchase of farm assets including farmland, machinery, and equipment such that the debt-to-farm-asset ratio after the purchase is less than or equal to ρ , where $0 < \rho < 1$

$$\frac{-[A_t - (P_t + \kappa + tc_p)x_t]}{(P_t + \kappa - tc_s)(L_t + x_t)} \leq \rho$$

Because the farmland price is stochastic, it is possible that the debt-to-farm-asset ratio can be above ρ , but we assume that the lender does not require sale of land in order to maintain this ratio. Substituting $A_t \equiv W_t - (P_t + \kappa - tc_s)L_t$ and rearranging terms yields

$$x_t \leq \max \left\{ 0, \frac{W_t - (1 - \rho)(P_t + \kappa - tc_s)L_t}{(P_t + \kappa + tc_p) - \rho(P_t + \kappa - tc_s)} \right\} \quad (17)$$

The constraints on control variable x_t are represented by Equation (7), where X_t is the set of all levels of x_t that satisfy Equations (14)–(17).

There are also constraints on investment in the mutual fund (x_{mt}). As described above, the amount invested in the mutual fund can be decided each year, and the control variable x_{mt} fully defines the level of the mutual fund asset. Since the investment in the mutual fund cannot be negative, we have

$$x_{mt} \geq 0 \quad (18)$$

Since the farm manager can potentially use available debt finances based on the farmland investment decision, for given W_t , P_t , L_t , and $x_t \in X_t$, the constraint on mutual fund investment x_{mt} is given by

$$\frac{-[A_t - (P_t + \kappa + tc)x_t - x_{mt}]}{(P_t + \kappa - tc_s)(L_t + x_t)} \leq \rho$$

Substituting $A_t \equiv W_t - (P_t + \kappa - tc_s)L_t$ and rearranging terms yields

$$x_{mt} \leq \max \{0, [W_t - (P_t + \kappa - tc_s)L_t - (P_t + \kappa + tc)x_t + \rho(P_t + \kappa - tc_s)(L_t + x_t)]\} \quad (19)$$

The constraints on control variable x_{mt} are represented by Equation (8), where X_{mt} is the set of all levels of x_{mt} that satisfy Equations (18) and (19).

To summarize, this investment problem has four state variables: gross return per acre (R_t), farmland price (P_t), farmland acreage (L_t), and net wealth (W_t), and one additional variable, mutual fund return (M_t), which is stochastic but does not depend on prior values for any state variables. The problem has two control variables: the number of acres of farmland to purchase or sell (x_t) and mutual fund investment (x_{mt}). These implicitly also determine the amount of either debt financing or investment in the risk-free asset. As given in Equation (1), the farm manager's objective is to choose an optimal policy $\{x_t^*, x_{mt}^*\}_{t=0}^T$ that maximizes the expected utility of terminal net wealth subject to constraints given in Equations (2)–(8). Finally, this is a finite horizon problem. Varying the length of the planning horizon, T , permits explicit analysis of the effect of farmer's age on investment decisions.

The model specified above in Equations (1)–(8) includes risky farmland, risky non-farm assets (mutual fund), and a risk-free nonfarm asset with debt financing on farmland. To explore the difference in farmland investment and financial decisions when risky non-farm assets are not considered, we also specify a special case of the model that excludes mutual fund investments. This model is represented by Equations (1)–(3) and (5)–(7), with $x_{mt} = 0$. It has one control variable x_t and the same four state variables: R_t , P_t , L_t , and W_t . We present results for cases with and without the mutual fund.

Objective Function

The objective function in Equation (1) is the expected utility of terminal net wealth. Another commonly used objective function in dynamic investment models is the additive utility of consumption or profits. The terminal wealth objective function is suitable for this analysis due to its simplicity and the straightforward definition of risk aversion. Although this objective function ignores intermediate consumption, it is of interest because it represents a class of problems that actually exists and provides insight into the nature of the multiperiod portfolio problem in a simple framework (Elton and Gruber 1975, p. 86). Jeffrey and Eidman (1991, p. 198) state that “additive utility... is likely to be inappropriate” because the assumption of additive independence is unrealistic in many cases. They show that an objective function of terminal wealth can be used to determine efficient sets for all forms of stochastic dominance and can be used to address the issue of long-run risk. Featherstone et al (1990, p. 85) also discuss the comparison of these objective functions, and support the objective of maximizing utility of terminal wealth.

The power utility function has been widely used in applications

$$U(W) = \frac{W^{1-\theta} - 1}{1-\theta} \quad (20)$$

where $\theta \geq 0$. If $\theta = 0$, the utility function exhibits risk-neutral preferences. If $\theta > 0$, the preferences are risk averse. Arrow (1971) asserts that preferences of a typical decision

maker exhibit decreasing absolute risk aversion with essentially constant relative risk aversion. These characteristics are satisfied by this utility function, where absolute risk aversion is $\frac{\theta}{W}$ and relative risk aversion is θ . When $\theta = 0$ Equation (20) yields $U(W) = W - 1$, and its strategically equivalent form is $u(W) = W$. Thus, for the risk-neutral case, the objective function is to maximize terminal net wealth. However, when $\theta > 0$ this utility function is not defined for $W < 0$. In order to define utility function for all W , we specify it for the model as

$$\begin{aligned} \text{When } \theta = 0: \quad & u(W) = W, & \text{for all } W \\ \text{When } \theta > 0: \quad & u(W) = \begin{cases} U(W) & \text{if } W \geq b \\ \frac{U(b)}{b} W & \text{if } W \leq b \end{cases} \end{aligned} \quad (21)$$

where $U(W)$ is given in Equation (20) and $b > 0$. For $\theta > 0$ utility function in Equation (21) is defined for all W and is a continuous function, with a linear portion for $W \leq b$ that passes through the origin. This specification represents risk-neutral preferences for $W \leq b$, while maintaining risk-averse preferences of Equation (20) for $W \geq b$.⁴

ESTIMATION OF EQUATIONS AND PARAMETERS

Numeric solution of the above investment problem by DP requires estimates of the stochastic equations and parameters of the model. We base these estimates on land price and farm management data from Southwestern Minnesota and on publicly available data on stock market returns.

Data

Stochastic equations for gross returns from crops per acre (R_t), farmland price (P_t), and mutual fund returns (M_t), given in Equations (2)–(4), are estimated using time-series data for the years 1967–2007. Gross returns are calculated for farms in the Southwestern Minnesota Farm Business Management Association (SWMFBMA), as published in annual reports (Nordquist et al various years; Olson et al various years). SWMFBMA records indicate that 44% of total cropped area is used for corn, 44% for soybeans, and 12% for other crops (Olson 2000). For simplicity, we assume that only corn and soybeans are produced, and gross returns per acre are calculated for a 50–50 corn-soybean mix. Time-series data for Southwestern Minnesota farmland prices for 1967–92 are from the Minnesota Rural Real Estate Market (Schwab and Raup 1989; Brekke et al 1993), and for 1990–2007 are from Taff (2008).⁵ Gross returns and farmland prices are adjusted for inflation to year 2000 using the Implicit Price Deflator for gross national product (USDC 2008). For mutual fund returns, we use inflation-adjusted compound annual total return for the S&P 500 for years 1967–2007 reported in Ibbotson Associates (2008).

Estimation of the Gross Return Equation

The gross return state equation is specified as an autoregressive model using the Box-Jenkins method of modeling time-series data for forecasting (Box and Jenkins 1976). This is an iterative method, which consists of identification of the order of an autoregressive model, estimation, and a diagnostic test to ensure that the error term is a white noise

Table 2. Estimation of gross return equation: Dependent variable $\ln R_t$

Variable	Coefficient estimate	Standard error	<i>t</i> -statistic
Constant	1.511810**	0.656765	2.301904
$\ln R_{t-1}$	0.742391**	0.112674	6.588854
$\widehat{Var}[\varepsilon_{1t}] = 0.030186, R^2 = 0.533, \text{ adjusted } R^2 = 0.521$			

**Significant at 5% level on a two-tailed test with the critical value of 2.021.

process. Data are transformed, if necessary, and the order of autoregressive model is increased until the error term is a white noise process. This approach suggests a first-order autoregressive model, $AR(1)$, for the gross return equation, with data transformation in natural logarithms, as specified in Equation (2). The equation is estimated by ordinary least squares (OLS), and results are presented in Table 2. The Ljung-Box (1979) statistic indicates that the null hypothesis of white noise error term is maintained. Furthermore, the hypothesis of normality of the error term⁶ is also maintained based on the Bera-Jarque (1981) test. Results of both these tests are provided in the Appendix. The OLS estimator is consistent for an autoregressive equation if the error term is a white noise process (Hamilton 1994). Additionally, if the error term is normally distributed, Hamilton (1994) shows that the OLS estimator for the autoregressive model is the same as that obtained by conditional maximum likelihood estimation.

Estimation of the Farmland Price Equation

The farmland price state equation is based on a model proposed by Burt (1986). Table 3 presents OLS regression results for the following specification:

$$\ln P_t = a_0 + a_1 \ln P_{t-1} + a_2 \ln P_{t-2} + a_3 \ln R_{t-1} + \varepsilon_{2t} \quad (22)$$

This specification gives a stable model by the Ljung-Box test for white noise and Bera-Jarque test of normality of error term (see the Appendix).⁷ Equation (22) has two lags on

Table 3. Estimation of farmland price equation: Dependent variable $\ln P_t$

Regression results				Reduction of lagged variable	
Variable	Coefficient estimate	Standard error	<i>t</i> -statistic	Variable	Coefficient estimate
Constant	0.353963	0.435578	0.812630	Constant	0.215000
$\ln P_{t-1}$	1.495469**	0.126870	11.787418	$\ln P_{t-1}$	0.908361
$\ln P_{t-2}$	-0.646337**	0.122997	-5.254888		
$\ln R_{t-1}$	0.130772*	0.072914	1.793513	$\ln R_{t-1}$	0.079432
$\widehat{Var}[\varepsilon_{2t}] = 0.010068, R^2 = 0.929, \text{ adjusted } R^2 = 0.922$				$\widehat{Var}[\varepsilon_{2t}] = 0.017292$	

**Significant at 5% level on a two-tailed test with the critical value of 2.021.

*Significant at 5% level on a one-tailed test with the critical value of 1.684.

Table 4. Estimation of mutual fund return equation: Dependent variable $\ln M_t$

Variable	Coefficient estimate	Standard error	t-statistic
Constant	0.057757**	0.025634	2.253140
$\widehat{Var}[\varepsilon_{3t}] = 0.026941$			

**Significant at 5% level on a two-tailed test with the critical value of 2.021.

farmland price, which requires inclusion of an additional state variable in the DP model, but Burt and Taylor (1989) provide a statistical procedure for reducing a state variable in these circumstances. Table 3 also presents the estimates obtained from the Burt and Taylor procedure. The reduced equation is represented as $\ln P_t = \alpha_0 + \alpha_1 \ln P_{t-1} + \alpha_2 \ln R_{t-1} + \varepsilon_{2t}$, where $\alpha_0 = \frac{a_0}{1-a_2}$, $\alpha_1 = \frac{a_1}{1-a_2}$, $\alpha_2 = \frac{a_3}{1-a_2}$, and the variance of error term is adjusted by dividing it by $(1 - (a_2)^2)$. Equation (3) in the model is the reduced equation.

Estimation of the Mutual Fund Return Equation

The Box-Jenkins approach suggests the specification given in Equation (4) for the mutual fund return equation. Regression results for this equation are presented in Table 4. This specification gives a stable model by the Ljung-Box test for white noise and Bera-Jarque test of normality of the error term (see the Appendix). These results indicate that the mutual fund return is stochastic but has a distribution that is not conditional on its lagged value. Thus, it is not necessary to include mutual fund return as a state variable in the DP model.

Parameters of the Net Wealth Equation

Values for parameters of the net wealth equation are based on SWMFBMA data and an interview with Dale Nordquist.⁸ The value of machinery and equipment per acre (κ) is \$300. The transaction cost for purchase of farmland is 1%, and there is no transaction cost on purchasing farm machinery and equipment. The transaction cost is 6% for selling farmland and 7% for selling machinery and equipment. The maximum debt-to-asset ratio allowed is 70% for purchasing farmland, machinery, and equipment ($\rho = 0.7$). As described for gross returns, corn and soybeans are planted in equal proportion, so we calculate the cost of production per acre for the 50–50 corn-soybean mix using SWMFBMA data for the years 1983–2007, as published in annual reports (Nordquist et al various years; Olson et al various years). After adjusting for inflation, the average value of cost of production per acre over this period equals \$201. We assume an opportunity cost of \$30 per acre for the farmer's own labor and management costs. Thus, the total cost of production per acre is $c = \$231$. The interest rate is assumed to be 6% on borrowing and 3% on lending (risk-free investment).

DYNAMIC PROGRAMMING FORMULATION AND NUMERICAL METHODS

The investment problem specified in Equations (1)–(8) can be solved by using DP. In this section, we describe the DP formulation, numerical methods, state space, and other specifications for solving the problem.

Dynamic Programming Formulation

The recurrence relation, or Bellman equation, for the investment problem can be written as

$$V_t(R, P, L, W) = \max_{(x, x_m)} \{E[V_{t+1}(R_{t+1}, P_{t+1}, L_{t+1}, W_{t+1}) | R_t = R, P_t = P, L_t = L, W_t = W, x_t = x, x_{mt} = x_m]\} \quad (23)$$

subject to the constraints in Equations (2)–(8) with the following boundary condition:

$$V_{T+1}(R, P, L, W) = u(W) \quad (24)$$

In Equation (23), $V_t(R, P, L, W)$ is the value function defined for $t = 0, 1, \dots, T$, as

$$V_t(R, P, L, W) \equiv \max_{\{x_\tau, x_{m\tau}\}_{\tau=t}^T} E[u(W_{T+1}) | R_t = R, P_t = P, L_t = L, W_t = W] \quad (25)$$

For each t , the value function is defined and solved for all states, not just the realized state. The value function $V_t(R, P, L, W)$ for each t is found by backward induction. Starting with V_{T+1} defined in Equation (24), we find V_T by using Equation (23). Moving backward, this process is repeated to characterize the value function for each t . The optimal policy for any given R, P, L , and W at any time $t = 0, 1, \dots, T$ can be obtained by solving

$$\max_{(x, x_m)} E[V_{t+1}(R_{t+1}, P_{t+1}, L_{t+1}, W_{t+1}) | R_t = R, P_t = P, L_t = L, W_t = W, x_t = x, x_{mt} = x_m] \quad (26)$$

subject to the constraints in Equations (2)–(8).

Numerical Methods and Implementation

This model lacks a closed-form solution. Thus, the value functions must be numerically approximated by computational methods. An efficient computational method for a continuous-state DP model, especially with multidimensions, is approximation of value functions by interpolation or piecewise interpolation, as introduced by Bellman and Dreyfus (1962) and Bellman et al (1963). The interpolation approach for approximating value functions is referred to as the collocation method (Miranda and Fackler 2002). The collocation method approximates the value function for each t with a linear combination of N basis functions using N prescribed points of the domain, called the collocation nodes

$$V_t(R, P, L, W) \approx \sum_{i=1}^N c_{it} \phi_i(R, P, L, W) \quad (27)$$

where R, P, L , and W are state variables; $\{\phi_i(\cdot)\}_{i=1}^N$ are the basis functions; and $\{c_{it}\}_{i=1}^N$ are basis coefficients. These N coefficients are computed given the values of V_t at N selected nodes from the state space. Implementation of the collocation method requires specification of (1) the type of collocation basis functions and nodes, and the number of nodes, (2) an optimization method, and (3) a method for finding the expected value.

The choice of basis functions and nodes depends on the characteristics of the function to be approximated. Approximation theory suggests the use of Chebyshev polynomial basis functions with Chebyshev nodes for smooth functions, and spline basis functions for nonsmooth functions (Miranda and Fackler 2002, pp. 118–135). Approximation with spline basis functions is made piecewise and has narrow supports; thus, it avoids the typical oscillating behavior of polynomial interpolation (Santos 1999). Linear spline functions with uniform nodes turns perform well for our model, and this scheme is used for approximating the value function. Using more nodes may yield more accurate results, but at the cost of more computer storage capacity and execution time, which pose serious computational limitations, especially in large-scale models. For comparing alternative numerical method schemes, Lohano and King (2006) developed and illustrated an accuracy test based on Monte Carlo simulation principles. Based on the accuracy results, we selected seven nodes for gross return (R_t), seven nodes for farmland price (P_t), five nodes for farmland acreage (L_t), and 21 nodes for net wealth (W_t).

A variety of methods are available for numerical optimization, described in Miranda and Fackler (2002) and Judd (1998). The golden-section search, Nelder-Mead algorithm, and Newton-Raphson method are designed to find a local optimum. The grid search method specifies a grid of points in the interval and finds the optimum from those points (Judd 1998, p. 100). Though this method is slow, it is likely to give a near global optimum, since the grid is specified for the entire range of the interval.⁹

Using the grid search method for optimization, we discretize the control spaces X_t into 41 uniform levels and X_{mt} into 25 uniform levels, which gives 1,025 combinations. The optimal policy is found in two steps. First, for each element of X_t , we find optimal x_{mt} . Then, we find the optimal $\{x_t, x_{mt}\}$. In the model without nonfarm assets ($x_{mt} = 0$), there is only one control variable x_t . For this model, we considered one-stage and two-stage grid search methods of optimization. Accuracy test results reported by Lohano and King (2006) suggest that the two-stage method is more efficient. Thus we use a two-stage method, where the control space X_t is discretized in 41 and 21 levels for the first and second stage, respectively.

Gauss-Hermite quadrature is used for numerical integration to evaluate the expected value function in the Bellman equation. For the model with mutual fund investment (risky nonfarm assets), we use three nodes for each of three stochastic error terms for evaluating the expected value function. This gives 27 combinations with their probabilities. In the model without mutual fund investment, we have two stochastic error terms, for each of which we use five nodes for the numerical integration. This gives 25 combinations with their probabilities.

State Space

The value function is approximated for the following ranges of the state variables: $230 \leq R_t \leq 540$; $1,010 \leq P_t \leq 2,840$; $400 \leq L_t \leq 2,000$; and $0 \leq W_t \leq 6,000,000$. Ranges for R_t and P_t are based on a 90% region for their distributions, estimated with their state equations. The range for farmland is chosen for fully owned farms. The lower bound of net wealth reflects the bankruptcy condition ($W_t < 0$), so $W_t \geq 0$ is the condition for staying in farming. The upper bound of net wealth is arbitrarily chosen to allow no debt at the upper bounds of farmland acreage and price. For the utility function in Equation (21), we set $b = 60,000$ considering the range of the net wealth from \$0 to \$6,000,000.

The value function is defined for the above state space, which includes the range of farmland as $400 \leq L_t \leq 2,000$. However, the farmer may choose to sell all farmland to exit farming, so the farmland in the next period will be 0. Therefore, for the backward recursion process, the value function also needs to be computed for $L_{t+1} = 0$. We assume that the business cannot re-enter farming after all farmland is sold. With no farmland, there can be no debt financing, since it is based on farmland as collateral. In this case, the investment problem is to choose an optimal portfolio of mutual fund and risk-free nonfarm assets, and the value function is $E[u(W_{T+1})]$ from the nonfarm asset portfolio. In the model without mutual fund investment, if the farmer chooses to sell all farmland and exit farming, net wealth W_{t+1} is all invested in the risk-free nonfarm asset. First, W_{T+1} is computed by compounding W_{t+1} at the return rate of the risk-free asset; then the utility function is evaluated as $u(W_{T+1})$. Since it is a risk-free investment, $u(W_{T+1})$ is the value function, as defined in Equation (25).

Net wealth in the next period (W_{t+1}) can also go outside of the state space due to the stochastic variables. Unlike farmland, it is generally not possible to use restrictions on the control variable to ensure that W_{t+1} is within the allowable range. When $W_{t+1} < 0$, the value function is computed for the bankruptcy condition. Under this condition, the farm is liquidated if net wealth is negative ($W_{t+1} < 0$) at any time, which makes $L_{t+1} = 0$. Thus, the value function can be computed for $L_{t+1} = 0$ as described above, but the compounding is done at the borrowing interest rate because $W_{t+1} < 0$. When W_{t+1} is greater than the upper bound, the extra amount above the upper bound earns returns from the portfolio of mutual fund and risk-free nonfarm assets. For the model without mutual fund investment, when W_{t+1} is greater than the upper bound, the extra amount above the upper bound earns the risk-free rate of return and is compounded to period $T + 1$ to compute the value function. Note that compounding of net wealth for $W_t = 0$ yields $W_{T+1} = 0$. In order to maintain continuity of the value function, the value function for zero current net wealth ($W_t = 0$) is set as $u(W_{T+1}) = u(0)$.

For gross return, to ensure that the state in period $t + 1$ is within the bounds, we assume $R_{t+1} = \max[\underline{R}, \min\{g(R_t, \varepsilon_{1,t+1}), \bar{R}\}]$, where $\underline{R}$ is the lower bound, $\bar{R}$ is the upper bound, and $g(R_t, \varepsilon_{1,t+1})$ denotes the right-hand side of Equation (2). This assumption assigns the probability mass for states outside the boundary to the value at the boundary. We also make this assumption for the farmland price state variable, P_{t+1} .

The DP model is programmed in MATLAB.¹⁰ For evaluating the basis functions and nodes for the collocation methods, and for numerical integration, we use MATLAB code developed by Miranda and Fackler (2002).

RESULTS FOR MODEL WITHOUT MUTUAL FUND

This section presents results for the model without mutual fund investment. We first present the optimal investment policy and stochastic dynamic path for the risk-neutral case ($\theta = 0$) and a 20-year planning horizon. Then we use comparative dynamics to compare the optimal decisions after changes in risk preferences, planning horizon, interest rate, and riskiness.

Optimal Investment Policy

The optimal policy for farmland purchase or sale (x_t^*) is a continuous function of the state variables—gross return (R_t), farmland price (P_t), farmland acreage (L_t), and net wealth

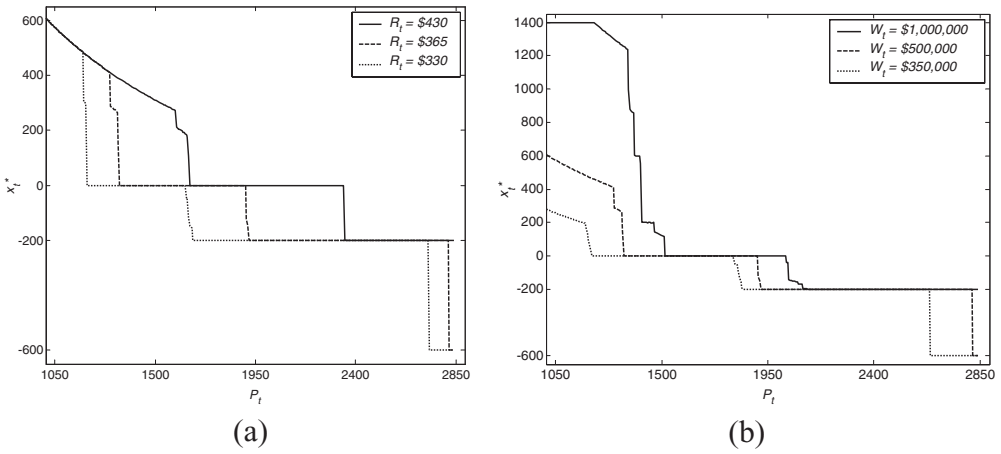


Figure 1. Optimal farmland investment policy as a function of farmland price for: (a) different levels of gross return, and (b) different levels of net wealth

(W_t)—for each time period t . We present here the optimal policy for selected levels of state variables for the initial year $t = 0$. Figure 1a shows the optimal farmland investment decision as a function of farmland price for three different levels of gross return, with fixed levels of farmland acreage (600 acres) and net wealth (\$500,000). When the gross return is \$365, the optimal investment decision is to purchase 606 acres if the farmland price is \$1,010. Optimal farmland purchase decreases as farmland price increases. For farmland prices from \$1,340 to \$1,905, the optimal choice is $x_t^* = 0$. For farmland prices greater than \$1,905, the optimal decision is to sell farmland. When we compare optimal decisions for different levels of gross return (Figure 1a), the optimal farmland investment is nondecreasing in gross return.

Figure 1b shows the optimal farmland investment decision as a function of farmland price for different levels of net wealth, with fixed levels of farmland acreage (600 acres) and gross return (\$365). The optimal farmland investment policy function is nonincreasing in farmland price and nondecreasing in net wealth. The optimal investment policy is nondecreasing in net wealth because of the lower opportunity cost of financing with a higher wealth level. This is due to the difference in interest rate on borrowing and lending. In addition, a wealthier farmer has access to more funds because his borrowing constraint is less binding. These results imply that farmland investment and disinvestment decisions are jointly determined by land prices, gross returns, and wealth, and are consistent with standard findings from investment theory.

In addition to positive and negative optimal investment policies (farmland purchase or sale) in Figure 1, $x_t^* = 0$ is optimal over a range of state variable values. This range of inaction is mainly due to the transaction costs on buying or selling farm assets, which include farmland, machinery, and equipment. The other reason for this is that when the debt-to-farm-asset ratio constraint is binding, this financial constraint prevents land purchase. There is also a range where it is optimal to sell 200 acres. Given the current farmland of 600 acres in this case, selling 200 acres lowers current farmland to 400 acres,

which is the minimum acreage required to stay in farming. At this level, the farm manager keeps open the option to remain in farming.

Stochastic Dynamic Path

Dynamic path analysis examines how state variables evolve over time starting from some initial state and following the optimal policy in each time period. When at least one variable is stochastic, the optimal policy can be determined for the initial state, but it is a contingency plan for each of the remaining periods. Optimal actions depend on the state, which will be known only after the realization of shocks in the stochastic variables. This means there will be a distribution of each state variable along the path, not an exact path. To explore the process over time, we use Monte Carlo simulation methods. We draw 500 trials for each of the 20 years for each of the error terms of the two stochastic variables—gross returns and farmland prices. Using these error terms and following the optimal policy in each time period yields 500 paths that represent the distribution of states for each year. The expected path is computed by taking their average for each year.

In the initial state at time $t = 0$, let the gross return be \$365 and farmland price be \$1,775. The farm manager has 600 acres of farmland and \$350,000 net wealth with 0.70 debt-to-farm-asset ratio (DFA_t), as defined in Equation (13). Table 5 presents simulation results from Years 1 to 10 for this initial state. The second and third columns show the trend in the expected net wealth and expected farm acreage. In Year 10, the expected net wealth is \$798,039 and the expected farm size is 621 acres. The probability of having exited farming reaches 11% by Year 10. The trend in the probability of exiting farming illustrates the cumulative effect of risk over the time. Table 5 also presents simulation results for a 600 acres farm with \$500,000 net wealth (0.57 debt-to-farm-asset ratio) and with \$1,000,000 net wealth (0.14 debt-to-farm-asset ratio) at the initial period. Results show that if the initial debt-to-farm-asset ratio is lower, the farmer has a lower risk of exiting farming and higher growth rate in expected farm size.

For illustration, Figure 2a presents a histogram of net wealth in Year 10 for the initial state of 600 acres of farmland with \$500,000 net wealth. It shows the cumulative effect of risk on wealth in 10 years and illustrates the possibility of bankruptcy (negative net wealth) as well as the possibility of accumulating considerable net wealth over time. Figure 2b presents a histogram of farm acreage in Year 10. The distribution of farm acreage is bimodal with peaks in probability mass at small and large farm sizes. It illustrates the cumulative effect of risk on farm growth with the possibility of exiting farming as well as the possibility of various farm sizes after 10 years.¹¹

Table 6 presents simulation results for probabilities of bankruptcy and voluntary exit from farming after 10 years for different levels of initial net wealth, keeping the same value of the other state variables. When the initial net wealth is \$350,000 (0.70 debt-to-farm-asset ratio), the probability of bankruptcy in 10 years is 4%, and that of voluntary exit is 7%, so the total probability of exit is 11%. Furthermore, there is a 13% probability of the debt-to-farm-asset ratio being greater than 0.7 and less than 1 if farming continues. Results in Table 6 show that if the initial debt-to-farm-asset ratio is lower, the farmer faces lower risks of bankruptcy, exiting farming, and financial stress.

Table 5. Simulation results of expected path and probabilities of exiting farming

Year <i>t</i>	Initial <i>DFA</i> = 0.7				Initial <i>DFA</i> = 0.57				Initial <i>DFA</i> = 0.14			
	Expected net wealth (\$)	Expected farm size (acres)	Prob. of exiting farming	Expected net wealth (\$)	Expected farm size (acres)	Prob. of exiting farming	Expected net wealth (\$)	Expected farm size (acres)	Expected net wealth (\$)	Expected farm size (acres)	Prob. of exiting farming	Prob. of exiting farming
0	350,000	600	–	500,000	600	–	1,000,000	600	1,000,000	600	–	–
1	383,149	600	0	542,149	600	0	1,072,149	600	1,072,149	600	0	0
2	422,552	505	0.002	592,712	531	0	1,154,954	703	1,154,954	703	0	0
3	457,557	486	0.008	639,177	534	0	1,238,124	815	1,238,124	815	0	0
4	489,642	491	0.016	685,655	570	0	1,332,067	925	1,332,067	925	0	0
5	528,744	501	0.032	742,705	607	0	1,449,100	1006	1,449,100	1006	0	0
6	574,456	523	0.046	806,666	664	0	1,573,911	1086	1,573,911	1086	0	0
7	623,834	554	0.062	879,078	720	0	1,711,147	1124	1,711,147	1124	0	0
8	674,050	592	0.078	954,513	776	0.008	1,851,038	1177	1,851,038	1177	0	0
9	733,501	617	0.090	1,041,528	811	0.010	2,003,273	1210	2,003,273	1210	0	0
10	798,039	621	0.110	1,132,596	831	0.016	2,149,728	1234	2,149,728	1234	0.002	0.002

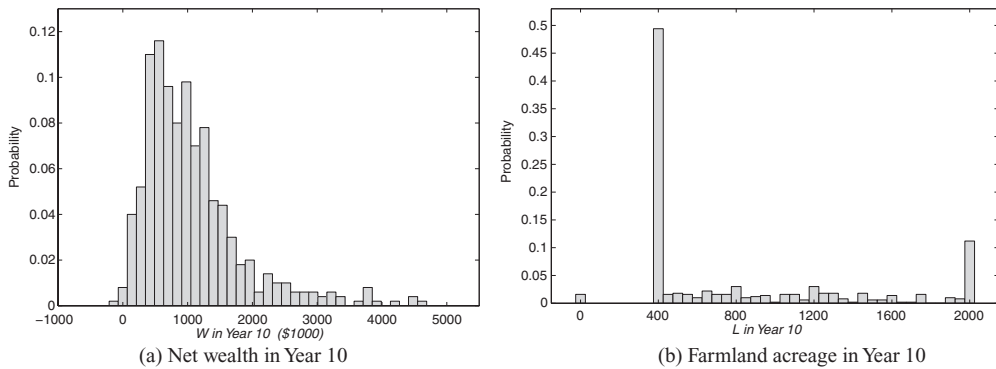


Figure 2. Distributions of net wealth and farmland acreage in Year 10

Table 6. Simulation results of probabilities of financial position in 10 years

Initial net wealth (\$) (Year 0)	Initial debt-to-farm-asset ratio (Year 0)	Prob. of bankruptcy (Year 10)	Prob. of exiting by choice (Year 10)	Combined prob. of exiting (Year 10)	Prob. of debt-to-farm-asset ratio between 0.7 and 1 (Year 10)
350,000	0.70	0.040	0.070	0.110	0.130
400,000	0.66	0.026	0.032	0.058	0.102
500,000	0.57	0.002	0.014	0.016	0.067
600,000	0.49	0	0.008	0.008	0.057
800,000	0.32	0	0.002	0.002	0.048
1,000,000	0.14	0	0.002	0.002	0.022

Comparative Dynamics

In this section we investigate how optimal decisions are affected by changes in key parameter values of the model including interest rate, planning horizon, riskiness, and risk aversion. In the results presented above, the interest rate on borrowed funds is assumed to be 6%. To explore the effect of a change in the borrowing interest rate, the model was also solved for 7%. All other parameters, including the interest rate on risk-free investment (3%), are unchanged. Figure 3a compares farmland investment policy with different interest rates on borrowing for particular levels of gross return (\$365), farmland acreage (600 acres), and net wealth (\$500,000). As expected, across the entire range of farmland prices, the optimal investment policy with a 7% borrowing interest rate is always less than or equal to that with 6%. Figure 3b compares the expected path of farm acreage for the initial state of gross return (\$365), farmland price (\$1,775), farmland (600 acres), and net wealth (\$1,000,000). Expected acreage in Year 10 is 1,118 acres with 7% borrowing interest rate, and 1,234 acres with 6%. Consistent with expectations, the farmer makes a lower investment in farmland when the borrowing interest rate rises.

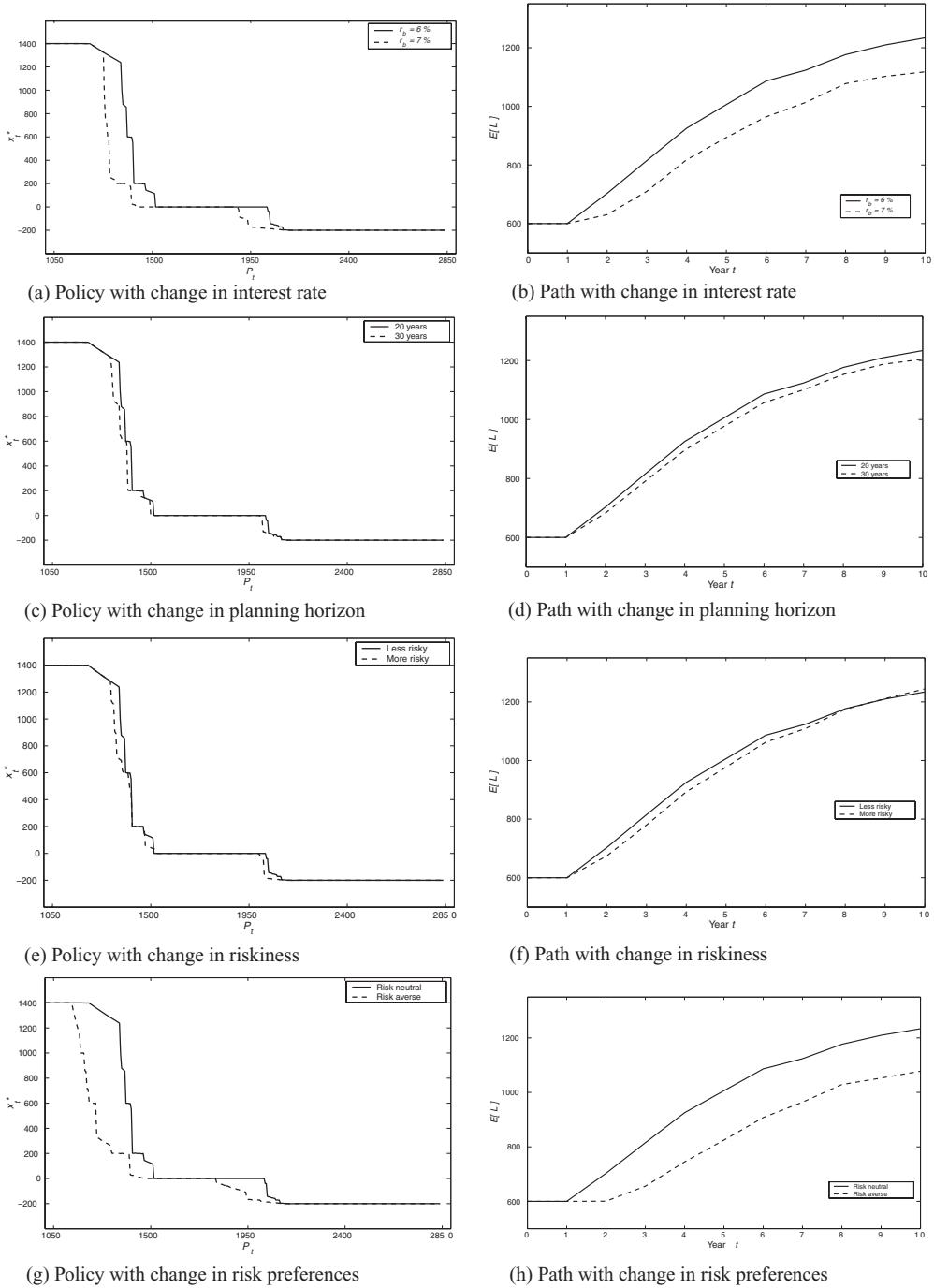


Figure 3. Optimal farmland investment policy and expected path of farmland: Comparative dynamics after changes in the interest rate, planning horizon, riskiness, and risk preferences

Figure 3c compares the optimal farmland investment policy with 20-year and 30-year planning horizons. Expected paths for farmland acreage are presented in Figure 3d. A farmer with a longer planning horizon tends to invest less in the risky asset. This reflects risk-avoiding behavior associated with the planning horizon.¹² Gross return and farmland price are stochastic state variables in the model. The variances of their error terms were estimated with their state equations. To explore the effect of a change in riskiness, the model is also solved after a 44% increase in the variances (or 20% increase in the standard deviations) of both error terms. Figure 3e shows that optimal farmland investment policy for higher variances is less than or equal to that for the lower variances across the range of farmland prices. As the gross return and farmland price have lognormal distributions, higher variances of error terms increase not only riskiness but also their expected values over time. Due to these two effects, the path of expected farmland is only slightly changed (Figure 3f).

Optimal decisions presented to this point are for the risk-neutral case with $\theta = 0$ in the utility function. To explore the effect of change in the risk preferences, the model is also solved for a risk-averse case with $\theta = 1$. Figure 3g shows that the optimal farmland investment for the risk-averse case is less than or equal to that for the risk-neutral case. Figure 3h shows that expected acreage along the optimal path is less for risk-averse preferences than for risk-neutral preferences.

The comparative dynamics results show that a risk-averse farmer makes a lower investment in risky farmland, a decision that reflects a risk-avoiding behavior. However, we also find that risk-avoiding behavior can also be attributed to other factors. With risk-neutral preferences, a change in the planning horizon affects the investment decisions. Furthermore, in contrast to a static model, a change in riskiness of returns and farmland prices also affects the investment decision even when the decision maker is risk neutral.

RESULTS FOR MODEL WITH MUTUAL FUND

This section presents the results for the complete model with mutual fund investment, as specified in Equations (1)–(8). This model includes risky farmland, mutual fund (risky nonfarm assets), and a risk-free nonfarm asset with debt financing on farmland. We also compare the results of the two models—with and without mutual fund investment—to investigate the difference in the decisions made by farm managers considering risky nonfarm assets and those not considering risky nonfarm assets. These models are solved for the risk-averse case with relative risk-aversion coefficient $\theta = 1$ and for 20-year planning horizon.

Optimal Investment Policy

Optimal investment decisions for models with and without mutual fund investment are presented in Table 7 for different levels of net wealth for fixed levels of gross return (\$365), farmland price (\$1,500), and farmland acreage (600 acres). The table also presents the optimal decisions for the farmland price of \$1,775. Results show that the optimal farmland investment policy is nonincreasing in farmland price and nondecreasing in net wealth in both models. However, when the farm manager has the opportunity to invest in the mutual fund, optimal farmland purchases are less than or equal to those in the model without mutual fund investment. For each level of net wealth, the initial

Table 7. Optimal investment policy with and without mutual fund

Current state		Model without mutual fund		Model with mutual fund		
Net wealth (\$)	Debt-to-farm-asset ratio	Optimal farmland transaction (acres)	Resulting debt-to-farm-asset ratio	Optimal farmland transaction (acres)	Optimal mutual fund investment (\$)	Resulting debt-to-farm-asset ratio
400,000	0.61	−200	For current farmland price of \$1,500	−200	49,330	0.48
600,000	0.41	0		0	209,652	0.61
1,000,000	0.01	0		0	492,986	0.50
400,000	0.66	−200	For current farmland price of \$1,775	−600	400,000	0 (exit farming)
600,000	0.49	−200		−200	137,362	0.41
1,000,000	0.14	0		0	568,269	0.63

Table 8. Expected path and probabilities with and without mutual fund

Year t	Model without mutual fund				Model with mutual fund			
	Expected net wealth (\$)	St. dev. of net wealth (\$)	Expected farmland (acres)	Prob. of exiting farming	Expected net wealth (\$)	St. dev. of net wealth (\$)	Expected farmland (acres)	Prob. of exiting farming
0	1,000,000	—	600	—	1,000,000	—	600	—
1	1,072,149	132,798	600	0	1,072,878	186,484	600	0
2	1,154,438	204,733	601	0	1,161,227	304,738	637	0
3	1,235,592	280,734	656	0	1,265,896	446,749	706	0.016
4	1,321,789	355,521	745	0	1,379,882	579,524	784	0.044
5	1,428,088	443,956	825	0	1,527,719	741,467	823	0.060
6	1,543,166	535,658	908	0	1,674,010	873,174	880	0.078
7	1,676,064	640,210	964	0	1,863,012	1,058,061	888	0.120
8	1,809,968	734,337	1,029	0.002	2,044,560	1,207,439	909	0.146
9	1,954,389	840,192	1,053	0.004	2,239,307	1,439,355	904	0.158
10	2,102,173	948,993	1,078	0.010	2,454,178	1,637,571	856	0.180

debt-to-farm-asset ratio is reported in the second column of Table 7. Debt-to-farm-asset ratios resulting from optimal investment decisions are also reported for both models.¹³ The results show that higher debt financing on farmland is optimal in the model with mutual fund.

Stochastic Dynamic Path

We use Monte Carlo simulation methods to explore the stochastic dynamic path using the methods described earlier. Table 8 presents results of path analysis for the models with and without mutual fund investment. In the initial state at time $t = 0$, the gross return is \$365 and the farmland price is \$1,775; the farm manager has 600 acres of farmland and \$1,000,000 net wealth. Expected acreage in Year 10 is 1,078 acres in the model without mutual fund investment and 856 acres in the model with mutual fund. The optimal portfolio has lower overall farmland acreage to accommodate mutual fund investment. The expected path of net wealth is higher in the model with mutual fund investment, however, the standard deviation of net wealth along the path is also higher. Table 8 also presents the probability of exiting farming, which is always voluntary in this case since the probability of bankruptcy is zero along the path for the given initial state. The probability of exiting farming in 10 years is 1% for the model without mutual fund investment, however, it is 18% for the model with mutual fund investment. Thus, the probability of exiting farming increases when mutual fund investment is permitted.

CONCLUSIONS

This study shows that optimal farmland investment decisions are dynamic in nature due to uncertainty, partial irreversibility, and the option to wait. Results of this study show ranges of inaction—the states where the optimal policy is to wait, neither purchasing nor selling farmland. Results show that a risk-averse farmer makes a lower investment in risky farmland reflecting risk-avoiding behavior, but we find that risk-avoiding behavior can also be attributed to lengthening of the planning horizon. Even when the decision maker is risk neutral, a change in riskiness of returns and farmland prices affects investment decisions.

The results show that allowing mutual fund investment increases both the expected value and variance of wealth. This indicates that the farmers investing in risky nonfarm assets realize higher returns but with higher risk. Thus, decisions about the level of non-farm investment depend mainly on the degree of risk aversion of the farmer. Furthermore, higher debt financing on farmland is optimal when mutual fund investment is allowed. However, we find that optimal farmland investment levels are generally lower and that the probability of exiting farming increases when there is a mutual fund investment opportunity.

NOTES

¹In effect, this treats the mutual fund as a liquid asset with risky returns. With costless adjustment, it is as though mutual fund balances are sold at the end of each year after returns have been realized and the proceeds are made available, along with other cash balances, for investment the following year. This eliminates the need for a state variable describing the current investment in the mutual fund.

²The purchase or sale decision for farmland is made at the beginning of each year t , so total land available for farming is L_{t+1} .

³In the model cash balance (A_t) is not included as a separate state variable because of the identity in Equation (11), as we can calculate it as $A_t \equiv W_t - (P_t + \kappa - tc_s)L_t$ for given values of the state variables and parameters.

⁴Featherstone et al (1990) also address the problem of defining the utility function for negative wealth. Their specification, however, gives a noncontinuous function. We also considered another specification offered by Keeney and Raiffa (1993, p. 173). However, their specification does not exhibit constant relative risk aversion.

⁵The first series (1967–92) and second series (1990–2007) of farmland prices are available for Southwestern Minnesota. Following Intriligator (1978), we spliced the two series in the scale of the second series (that are more recent data) by multiplying the first series by 0.868, which is the ratio of the average price of the second series and that of the first series during the overlap years 1990–92. Furthermore, we find that the ratio of two prices is also around 0.868 at each of the years 1990–92.

⁶The distribution of the error term is also used in solving the DP problem numerically.

⁷We also check for a statistical relationship between ε_{1t} and ε_{2t} , that is, between error term of gross return equation and error term of farmland price equation. The covariance of their residuals is computed as 0.000886, and the correlation is 0.054. Since the covariance is close to zero, the OLS estimates are efficient.

⁸Associate Director, Center for Farm Financial Management, Department of Applied Economics, University of Minnesota, St. Paul. Date of interview is April 6, 2009.

⁹Use of grid search method with the collocation method is referred to as hybrid method by Fackler and Miranda (1999), because this method discretizes the control space while using continuous method for approximating the value function.

¹⁰MATLAB code is available online at: <http://www.apec.umn.edu/faculty/rking/research.html>.

¹¹Note that the distribution of the error terms is across time, and the results are presented for a single farm with some initial state.

¹²There is also an end period effect on policy. The analysis of the optimal policy, not presented here, indicates that farmland purchases are much lower during the last five years near the end of planning horizon. Furthermore, there is more probability of selling some or all land for earlier retirement near the end of planning horizon.

¹³As defined in Equation (13), debt-to-farm-asset ratio is $DFA_t = \max\{0, \frac{-A_t}{(P_t + \kappa - tc_s)L_t}\}$. Debt-to-farm-asset ratio resulting from optimal investment decisions is $DFA_t^* = \max\{0, \frac{[-A_t - (P_t + \kappa + tc)x_t^*]}{(P_t + \kappa - tc_s)(L_t + x_t^*)}\}$ in the model without mutual fund, and it is $DFA_t^{**} = \max\{0, \frac{[-A_t - (P_t + \kappa + tc)x_t^* - x_{mt}^*]}{(P_t + \kappa - tc_s)(L_t + x_t^*)}\}$ in the model with mutual fund.

REFERENCES

- Arrow, K. J. 1971.** *Essays in the Theory of Risk Bearing*. Chicago, IL: Markham Publishing Company.
- Baker, C. B. 1977.** Introduction to economic growth of the agricultural firm. In *Economic growth of the agricultural firm*, pp. 1–7. Technical Bulletin 86. Pullman, WA: College of Agriculture Research Center, Washington State University, February.
- Bellman R. E. and S. E. Dreyfus. 1962.** *Applied Dynamic Programming*. Princeton, NJ: Princeton University Press.
- Bellman, R. E., R. Kalaba and B. Kotkin. 1963.** Polynomial approximation—A new computational technique in dynamic programming: Allocation processes. *Mathematics of Computation* 17: 155–61.
- Bera, A. and C. Jarque. 1981.** Efficient tests for normality, heteroscedasticity, and serial independence of regression residuals: Monte Carlo evidence. *Economics Letters* 7: 313–18.
- Box, G. E. P. and G. M. Jenkins. 1976.** *Time Series Analysis: Forecasting and Control*. Revised Edition. San Francisco, CA: Holden-Day.

- Brake, J. R. 1983.** Financial crisis in agriculture: Discussion. *American Journal of Agricultural Economics* 65: 953–54.
- Brekke, J., H.L. Tao and P. M. Raup. 1993.** The Minnesota rural real estate market in 1992. Economic Report ER 93-5. St. Paul, Minnesota: University of Minnesota, July.
- Burt, O. R. 1986.** Econometric modeling of the capitalization formula of farmland prices. *American Journal of Agricultural Economics* 68: 10–26.
- Burt, O. R. and C. R. Taylor. 1989.** Reduction of state variable dimension in stochastic dynamic optimization models which use time-series data. *Western Journal of Agricultural Economics* 14 (2): 213–22.
- Chavas, J. P. and A. Thomas. 1999.** A dynamic analysis of land prices. *American Journal of Agricultural Economics* 81: 772–84.
- Collins, R. A. and L. S. Karp. 1993.** Lifetime leverage choice for proprietary farmers in a dynamic stochastic environment. *Journal of Agricultural and Resource Economics* 18 (2): 225–38.
- Dixit, A. K. and R. S. Pindyck. 1994.** *Investment under Uncertainty*. Princeton, NJ: Princeton University Press.
- Elton, E. J. and M. J. Gruber. 1975.** *Finance as a Dynamic Process*. Englewood Cliffs, NJ: Prentice-Hall, Inc.
- Fackler, P. L. and M. J. Miranda. 1999.** Hybrid methods for continuous state dynamic programming. Presented at Annual Conference of the Society for Computational Economics, Boston, Massachusetts, June 24–26.
- Featherstone, A. M., P. V. Preckel and T. G. Baker. 1990.** Modeling farm financial decisions in dynamic and stochastic environment. *Agricultural Finance Review* 50: 80–99.
- Hamilton, J. D. 1994.** *Time Series Analysis*. Princeton, NJ: Princeton University Press.
- Hirshleifer, J. 1970.** *Investment, Interest, and Capital*. Englewood Cliffs, NJ: Prentice-Hall, Inc.
- Ibbotson Associates. 2008.** *Stocks, Bonds, Bills, and Inflation 2008 Yearbook*. Chicago, IL: Ibbotson Associates.
- Intriligator, M. D. 1978.** *Econometric Models, Techniques, & Applications*. Englewood Cliffs, NJ: Prentice-Hall.
- Jeffrey, S. R. and V. R. Eidman. 1991.** The application of multivariate stochastic dominance to agricultural economic problems. *Canadian Journal of Agricultural Economics* 39: 193–209.
- Jensen, F. E. and L. N. Langemeier. 1996.** Optimal leverage with risk aversion: Empirical evidence. *Agricultural Finance Review* 56: 85–97.
- Judd, K. L. 1998.** *Numerical Methods in Economics*. Cambridge, MA: The MIT Press.
- Keeney, R. L. and H. Raiffa. 1993.** *Decisions with Multiple Objectives: Preferences and Value Trade-offs*. New York, NY: Cambridge University Press.
- Kocherlakota, N. R. 1996.** The equity premium: It's still a puzzle. *Journal of Economic Literature* 34: 42–71.
- Larson, D. K., M. S. Stauber and O. R. Burt. 1974.** Economic analysis of farm firm growth in Northcentral Montana. Research Report 62. Bozeman: Montana Agricultural Experiment Station, Montana State University, August.
- Ljung, G. and G. Box. 1979.** On a measure of lack of fit in time series models. *Biometrika* 66: 265–70.
- Lohano, H. D. and R. P. King. 2006.** Accuracy of numerical solution to dynamic programming models. Staff Paper P06-3, Staff Paper Series. St. Paul: Department of Applied Economics, University of Minnesota, January.
- Minden, A. J. 1968.** Dynamic programming: A tool for farm firm growth research. *Canadian Journal of Agricultural Economics* 16: 38–45.
- Miranda, M. J. and P. L. Fackler. 2002.** *Applied Computational Economics and Finance*. Cambridge, MA: The MIT Press.

- Nemhauser, G. L. 1966.** *Introduction to Dynamic Programming*. New York, NY: John Wiley & Sons.
- Nordquist, D. and Others. Various years.** Annual report of the Southwestern Minnesota Farm Business Management Association. Staff Paper Series. St. Paul: Department of Applied Economics, University of Minnesota.
- Olson, K. D. 2000.** Crop mix changes over time in the SW Association. Summary table based on data collected from the Southwestern Minnesota Farm Business Management Association database, June.
- Olson, K. D. and Others. Various years.** Annual report of the Southwestern Minnesota Farm Business Management Association. Staff Paper Series. St. Paul: Department of Applied Economics, University of Minnesota.
- Pederson, G. D. and J. R. Brake. 1982.** The distribution of financial impacts of alternative agricultural credit policies. Research Report 424. East Lansing: The Michigan State University Agricultural Experiment Station, February.
- Puterman, M. L. 1989.** *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. New York, NY: John Wiley.
- Rust, J. 1996.** Numerical dynamic programming in economics. In *Handbook of Computational Economics*, Volume I, edited by H. M. Amman, D. A. Kendrick and J. Rust, pp. 619–29. Amsterdam: Elsevier Science B.V.
- Santos, M. S. 1999.** Numerical solution of dynamic economic models. In *Handbook of Macroeconomics*, edited by J. B. Taylor and M. Woodford, pp. 305–80. Amsterdam: Elsevier Science B.V.
- Schnitkey, G. D., C. R. Taylor and P. J. Barry. 1989.** Evaluating farmland investments considering dynamic stochastic returns and farmland prices. *Western Journal of Agricultural Economics* 14 (1): 143–56.
- Schwab, A. and P. M. Raup. 1989.** The Minnesota rural real estate market in 1988. Economic Report ER 89-3. St. Paul, Minnesota: University of Minnesota, July.
- Taff, S. J. 2008.** Minnesota farm real estate sales: 1990–2007. Staff Paper P08-5, Staff Paper Series. St. Paul: Department of Applied Economics, University of Minnesota, June.
- U.S. Department of Agriculture (USDA). 2008.** Quick Stats: Agricultural statistics data base. USDA National Agricultural Statistics Service, <http://www.nass.usda.gov/QuickStats/> (accessed September 14, 2008).
- U.S. Department of Commerce (USDC). 2008.** National income and product accounts table. National Economic Accounts, Bureau of Economic Analysis, <http://www.bea.gov/bea/dn/nipaweb/SelectTable.asp> (accessed September 14, 2008).

APPENDIX

Table A1. Ljung-Box test for white noise error term

K	Error term from gross return equation		Error term from farmland price equation		Error term from mutual fund return equation	
	$Q(K)$ statistic	p-value	$Q(K)$ statistic	p-value	$Q(K)$ statistic	p-value
1	0.0323	0.857	0.0972	0.755	0.3606	0.548
2	0.3102	0.856	0.1030	0.950	1.6993	0.428
3	0.3103	0.958	0.1099	0.991	1.7199	0.633
4	0.3146	0.989	0.1199	0.998	2.8456	0.584
5	0.9427	0.967	1.9411	0.857	2.9843	0.702

(Continued)

Table A1. Continued

<i>K</i>	Error term from gross return equation		Error term from farmland price equation		Error term from mutual fund return equation	
	<i>Q(K)</i> statistic	<i>p</i> -value	<i>Q(K)</i> statistic	<i>p</i> -value	<i>Q(K)</i> statistic	<i>p</i> -value
6	1.0943	0.982	4.5389	0.604	3.4101	0.756
7	2.1323	0.952	4.5964	0.709	3.4186	0.844
8	2.2221	0.973	4.7018	0.789	4.1171	0.846
9	4.9751	0.836	5.5017	0.789	4.1268	0.903
10	5.3235	0.869	6.9592	0.729	4.1350	0.941
11	6.7624	0.818	8.9298	0.628	4.5967	0.949
12	7.0573	0.854	9.2307	0.683	4.6080	0.970
13	8.0084	0.843	9.2813	0.751	4.7222	0.981
14	8.8712	0.839	10.805	0.701	5.0012	0.986
15	9.0422	0.875	10.876	0.761	5.2614	0.990

Table A2. Bera-Jarque test for normality of error terms

Error term from gross return equation		Error term from farmland price equation		Error term from mutual fund return equation	
<i>BJ</i> statistic	<i>p</i> -value	<i>BJ</i> statistic	<i>p</i> -value	<i>BJ</i> statistic	<i>p</i> -value
3.279623	0.194017	0.047132	0.976710	5.087505	0.078571